Face Recognition using Eigenanalysis

Eigenvalues and eigenvectors as directions and amounts of scatter

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Abstract. We apply eigenanalysis to the recognition of faces. Each 200×180 photograph is treated as a vector in $\mathbb{R}^{36000}$. After the mean face is subtracted, the directions in which the images vary most (the *eigenfaces*) and the variance along each are given by the eigenvectors and eigenvalues of the scatter matrix. This matrix is too large to decompose directly, so we work with the small Gram matrix $\mathbf{A}^\top\mathbf{A}$ and prove that it carries the same information. On 50 photographs of 10 people, with the eigenfaces computed from four photographs per person, nine eigenfaces retain 96.9% of the training variance and a nearest-neighbour rule identified all ten held-out faces. The test set is small, and we state what it does and does not show.

1 Introduction

A grayscale photograph of 200×180 pixels is a list of 36 000 numbers. Comparing two photographs number by number is slow, and a small change in lighting alters every entry although the face is the same. Faces are nevertheless far from arbitrary arrays: neighbouring pixels are strongly correlated and the features of a face lie in similar places from one person to the next. One therefore expects the photographs to vary in only a few essential ways, and a description that records just those ways should be much shorter than the raw image.

The eigenface method of Sirovich and Kirby and of Turk and Pentland makes this precise. The images are regarded as a sample from a distribution on $\mathbb{R}^d$, where d is the number of pixels. The directions of greatest spread are the eigenvectors of the sample covariance matrix, and the eigenvalues are the variances along them. Keeping a few leading directions gives each face a short coordinate vector, and recognition is carried out on these coordinates. In modern terms this is principal component analysis applied to images.

Section 2 develops the mathematics and Section 3 checks it by hand on a three-pixel example. Section 4 reports an experiment on 50 photographs, Section 5 records the practical difficulties, and Section 6 discusses limitations. Figure 1 gives an overview.

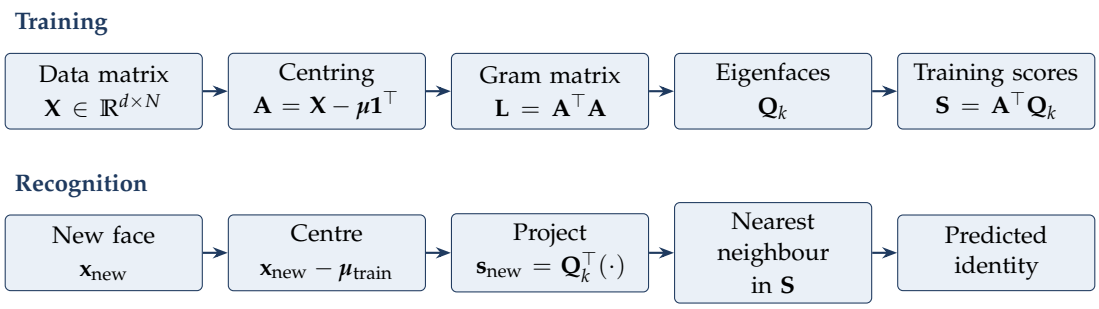


Figure 1. Outline of the method. Everything in the top row is computed once from the training images; a new face is centred with the *training* mean and projected on the same eigenfaces.

2 Mathematical framework

2.1 Centred data and variance along a direction

Reading a photograph of $h = 200$ rows and $w = 180$ columns row by row gives a vector $\mathbf{x} \in \mathbb{R}^d$ with $d = hw = 36\,000$ and entries in $[0, 255]$. The N photographs form the columns of $\mathbf{X} \in \mathbb{R}^{d \times N}$. Let $\boldsymbol{\mu} = \frac{1}{N}\mathbf{X}\mathbf{1}$ be the mean face, where $\mathbf{1}$ is the vector of ones, and let

$$\mathbf{A} = \mathbf{X} - \boldsymbol{\mu}\mathbf{1}^\top = (\mathbf{a}_1 \ \cdots \ \mathbf{a}_N), \qquad \mathbf{a}_i = \mathbf{x}_i - \boldsymbol{\mu},$$

be the centred data.

Lemma 1. $\mathbf{A}\mathbf{1} = \mathbf{0}$; *hence* $\mathbf{a}_1 + \cdots + \mathbf{a}_N = \mathbf{0}$ *and* $\operatorname{rank}(\mathbf{A}) \le N - 1$.

Proof. $\mathbf{A}\mathbf{1} = \mathbf{X}\mathbf{1} - \boldsymbol{\mu}\,\mathbf{1}^\top\mathbf{1} = N\boldsymbol{\mu} - N\boldsymbol{\mu} = \mathbf{0}$, so the columns of $\mathbf{A}$ are linearly dependent. □

For a unit vector $\mathbf{u} \in \mathbb{R}^d$, the coordinates $\mathbf{u}^\top\mathbf{a}_i$ of the centred faces along $\mathbf{u}$ have mean zero by Lemma 1, and their sample variance is

$$\operatorname{Var}(\mathbf{A}^\top\mathbf{u}) = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{u}^\top\mathbf{a}_i)^2 = \mathbf{u}^\top\mathbf{C}\,\mathbf{u}, \qquad \mathbf{C} = \frac{1}{N}\mathbf{M}, \quad \mathbf{M} = \mathbf{A}\mathbf{A}^\top. \tag{1}$$

The matrices $\mathbf{C}$ and $\mathbf{M}$ are the sample covariance and the scatter matrix; both are symmetric and positive semidefinite, since $\mathbf{u}^\top\mathbf{M}\mathbf{u} = \|\mathbf{A}^\top\mathbf{u}\|^2 \ge 0$.

Theorem 2 (Directions of maximal variance)**.** *Let* $\mathbf{M}$ *be symmetric positive semidefinite with eigenvalues* $\lambda_1 \ge \cdots \ge \lambda_d \ge 0$ *and orthonormal eigenvectors* $\mathbf{q}_1, \ldots, \mathbf{q}_d$*. Then* $\max_{\|\mathbf{u}\|=1} \mathbf{u}^\top\mathbf{M}\mathbf{u} = \lambda_1$*, attained at* $\mathbf{q}_1$*; and for* $j \ge 2$ *the maximum of* $\mathbf{u}^\top\mathbf{M}\mathbf{u}$ *over unit vectors orthogonal to* $\mathbf{q}_1, \ldots, \mathbf{q}_{j-1}$ *equals* λ_j*, attained at* $\mathbf{q}_j$*.*

Proof. Write $\mathbf{u} = \sum_i c_i\mathbf{q}_i$ with $\sum_i c_i^2 = 1$. Then $\mathbf{u}^\top\mathbf{M}\mathbf{u} = \sum_i \lambda_i c_i^2$, a weighted average of the eigenvalues with weights $c_i^2 \ge 0$ summing to one. It cannot exceed λ_1, and equals it at $\mathbf{u} = \mathbf{q}_1$. If $\mathbf{u} \perp \mathbf{q}_1, \ldots, \mathbf{q}_{j-1}$ then $c_1 = \cdots = c_{j-1} = 0$ and the same argument gives the bound λ_j. □

Since $\mathbf{C} = \mathbf{M}/N$, the first principal direction is the unit vector along which the faces vary most, with variance λ_1/N, and each further direction is the best choice orthogonal to those before it. Throughout, λ_i denotes an eigenvalue of $\mathbf{M}$ (equivalently of $\mathbf{L}$ below); the variance along the i-th direction is λ_i/N. (For instance, if centred two-dimensional data have covariance $\left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)$, the variances are 4 and 2 along $(1,1)/\sqrt{2}$ and $(1,-1)/\sqrt{2}$, and the first direction carries $4/6 \approx 67\%$ of the total.)

2.2 The Gram-matrix identity

For our data $\mathbf{M}$ is a $36\,000 \times 36\,000$ matrix with 1.296×10^9 entries, about 10.4 GB in double precision, and decomposing it directly is not feasible. The following result avoids this by using the $N \times N$ matrix $\mathbf{L} = \mathbf{A}^\top\mathbf{A}$, whose entries are $L_{ij} = \mathbf{a}_i^\top\mathbf{a}_j$.

Proposition 3. *Let* $\mathbf{A} \in \mathbb{R}^{d \times N}$ *have rank* r*. If* $\mathbf{v}_1, \ldots, \mathbf{v}_r$ *are orthonormal eigenvectors of* $\mathbf{L} = \mathbf{A}^\top\mathbf{A}$ *for its positive eigenvalues* $\lambda_1, \ldots, \lambda_r$*, then*

$$\mathbf{u}_i = \frac{\mathbf{A}\mathbf{v}_i}{\sqrt{\lambda_i}}, \qquad i = 1, \ldots, r,$$

are orthonormal eigenvectors of $\mathbf{M} = \mathbf{A}\mathbf{A}^\top$ *with the same eigenvalues. Moreover* $\mathbf{L}$ *and* $\mathbf{M}$ *have the same positive eigenvalues, and there are exactly* r *of them.*

Proof. From $\mathbf{L}\mathbf{v}_i = \lambda_i \mathbf{v}_i$ we get $\mathbf{M}(\mathbf{A}\mathbf{v}_i) = \mathbf{A}\mathbf{L}\mathbf{v}_i = \lambda_i \mathbf{A}\mathbf{v}_i$, and $\|\mathbf{A}\mathbf{v}_i\|^2 = \mathbf{v}_i^\top \mathbf{L}\mathbf{v}_i = \lambda_i > 0$. For $i, j \le r$, $\mathbf{u}_i^\top \mathbf{u}_j = \mathbf{v}_i^\top \mathbf{L}\mathbf{v}_j / \sqrt{\lambda_i \lambda_j} = \delta_{ij}$. Conversely, if $\mathbf{M}\mathbf{u} = \lambda \mathbf{u}$ with $\lambda > 0$, then $\mathbf{L}(\mathbf{A}^\top \mathbf{u}) = \lambda\, \mathbf{A}^\top \mathbf{u}$ and $\mathbf{A}^\top \mathbf{u} \neq \mathbf{0}$, so every positive eigenvalue of $\mathbf{M}$ is one of $\mathbf{L}$ with at least the same multiplicity, and the two spectra agree. The number of positive eigenvalues is $\operatorname{rank}(\mathbf{L}) = \operatorname{rank}(\mathbf{A}) = r$. □

The vectors $\mathbf{u}_i$ are the *eigenfaces*; reshaped to 200×180 they can be displayed as images. We write $\mathbf{Q} = (\mathbf{u}_1 \ \cdots \ \mathbf{u}_r)$, so that $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$. The cost drops from an eigenproblem of size d to one of size N (forming $\mathbf{L}$ takes $O(dN^2)$ operations), which here is negligible. Since $\mathbf{L}\mathbf{1} = \mathbf{A}^\top \mathbf{A}\mathbf{1} = \mathbf{0}$, the matrix $\mathbf{L}$ has eigenvalue 0 in exact arithmetic; in floating point it appears as a tiny number of either sign. We therefore keep only eigenvalues with $\lambda_i > 10^{-10}\lambda_1$, which gives the numerical rank r.

2.3 Number of components, scores and reconstruction

The i-th direction accounts for the fraction $\rho_i = \lambda_i / \sum_{j \le r} \lambda_j$ of the total variance, and the first k directions for $R_k = \sum_{i \le k} \rho_i$. We use

$$k = \min\{k : \ R_k \ge 0.95\}.$$

With $\mathbf{Q}_k = (\mathbf{u}_1 \ \cdots \ \mathbf{u}_k)$, the *score vector* of a face and its reconstruction are

$$\mathbf{s}_i = \mathbf{Q}_k^\top \mathbf{a}_i \in \mathbb{R}^k, \qquad \hat{\mathbf{x}}_i = \boldsymbol{\mu} + \mathbf{Q}_k \mathbf{s}_i, \qquad \mathbf{S} = \mathbf{A}^\top \mathbf{Q}_k \in \mathbb{R}^{N \times k},$$

where the i-th row of $\mathbf{S}$ is $\mathbf{s}_i^\top$.

Proposition 4. *Let $\mathbf{V}_k = (\mathbf{v}_1 \ \cdots \ \mathbf{v}_k)$ and $\boldsymbol{\Lambda}_k = \operatorname{diag}(\lambda_1, \ldots, \lambda_k)$. Then*

(a) $\mathbf{S} = \mathbf{V}_k \boldsymbol{\Lambda}_k^{1/2}$, so $\mathbf{S}^\top \mathbf{S} = \boldsymbol{\Lambda}_k$. Each column of $\mathbf{S}$ has zero sum, hence the score coordinates are uncorrelated and the j-th has variance λ_j / N.

(b) $\sum_{i=1}^N \|\mathbf{a}_i - \mathbf{Q}_k \mathbf{s}_i\|^2 = \sum_{j=k+1}^r \lambda_j$, so $1 - R_k$ is the fraction of the total squared deviation lost in reconstruction.

Proof. (a) The j-th column of $\mathbf{S}$ is $\mathbf{A}^\top \mathbf{u}_j = \mathbf{L}\mathbf{v}_j / \sqrt{\lambda_j} = \sqrt{\lambda_j}\, \mathbf{v}_j$, which gives $\mathbf{S} = \mathbf{V}_k \boldsymbol{\Lambda}_k^{1/2}$ and $\mathbf{S}^\top \mathbf{S} = \boldsymbol{\Lambda}_k$. Also $\mathbf{S}^\top \mathbf{1} = \mathbf{Q}_k^\top \mathbf{A}\mathbf{1} = \mathbf{0}$. (b) Each $\mathbf{a}_i$ lies in the column space of $\mathbf{A}$, spanned by $\mathbf{u}_1, \ldots, \mathbf{u}_r$, so $\mathbf{a}_i = \sum_{j \le r} s_{ij} \mathbf{u}_j$ and $\mathbf{a}_i - \mathbf{Q}_k \mathbf{s}_i = \sum_{j > k} s_{ij} \mathbf{u}_j$. Its squared norm is $\sum_{j>k} s_{ij}^2$; summing over i and using (a) gives $\sum_{j>k} \lambda_j$. The total is $\sum_i \|\mathbf{a}_i\|^2 = \operatorname{tr}(\mathbf{L}) = \sum_{j \le r} \lambda_j$. □

Part (b) refers to the images used to build the basis; it does not bound the reconstruction error of a new image.

2.4 Recognition

A new photograph $\mathbf{x}_{\text{new}}$ is centred with the mean of the *training* images, projected on the training eigenfaces, and assigned the identity of the nearest training face in Euclidean distance:

$$\mathbf{s}_{\text{new}} = \mathbf{Q}_k^\top (\mathbf{x}_{\text{new}} - \boldsymbol{\mu}_{\text{train}}), \qquad \hat{\imath} = \arg\min_i \|\mathbf{s}_i - \mathbf{s}_{\text{new}}\|.$$

Because $\mathbf{Q}_k$ has orthonormal columns, $\|\mathbf{Q}_k \mathbf{s} - \mathbf{Q}_k \mathbf{t}\| = \|\mathbf{s} - \mathbf{t}\|$, so this is the same as a nearest-neighbour search among the projected faces in pixel space, carried out in k dimensions instead of d.

3 A worked example

Take $d = 3$ pixels and $N = 3$ images, $\mathbf{x}_1 = (3,2,3)^\top$, $\mathbf{x}_2 = (1,3,2)^\top$, $\mathbf{x}_3 = (2,1,1)^\top$. The mean face is $\boldsymbol{\mu} = (2,2,2)^\top$ and

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \end{pmatrix}, \qquad \mathbf{L} = \mathbf{A}^\top\mathbf{A} = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} = 3\mathbf{I} - \mathbf{1}\mathbf{1}^\top.$$

The rows of $\mathbf{A}$ sum to zero, as Lemma 1 requires. Since $\mathbf{1}\mathbf{1}^\top$ has eigenvalues $3, 0, 0$, the eigenvalues of $\mathbf{L}$ are $3, 3, 0$, with $\operatorname{tr}(\mathbf{L}) = 6$. Take the orthonormal eigenvectors $\mathbf{v}_1 = (1,-1,0)^\top/\sqrt{2}$ and $\mathbf{v}_2 = (1,1,-2)^\top/\sqrt{6}$ for $\lambda = 3$; the eigenvector $\mathbf{1}/\sqrt{3}$ for $\lambda = 0$ gives $\mathbf{A}\mathbf{1} = \mathbf{0}$ and is discarded. Then

$$\mathbf{u}_1 = \frac{\mathbf{A}\mathbf{v}_1}{\sqrt{3}} = \frac{1}{\sqrt{6}}\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \qquad \mathbf{u}_2 = \frac{\mathbf{A}\mathbf{v}_2}{\sqrt{3}} = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \qquad \mathbf{u}_1^\top\mathbf{u}_2 = 0,$$

and one checks directly that $\mathbf{A}\mathbf{A}^\top\mathbf{u}_i = 3\mathbf{u}_i$. Because $\lambda_1 = \lambda_2$, only the plane spanned by $\mathbf{u}_1$ and $\mathbf{u}_2$ is determined by the data, so the individual eigenfaces depend on the choice of basis made above.

The scores on $\mathbf{u}_1$ are $\mathbf{s}_{\cdot 1} = (3/\sqrt{6},\ -3/\sqrt{6},\ 0)^\top \approx (1.225, -1.225, 0)^\top$, with sum of squares $3 = \lambda_1$, as Proposition 4(a) predicts. With $k = 1$,

$$\hat{\mathbf{x}}_1 = \boldsymbol{\mu} + \mathbf{u}_1 s_{11} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} + \frac{3}{\sqrt{6}} \cdot \frac{1}{\sqrt{6}}\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1.5 \\ 2.5 \end{pmatrix} \approx \mathbf{x}_1.$$

Likewise $\hat{\mathbf{x}}_2 = (1, 2.5, 1.5)^\top$ and $\hat{\mathbf{x}}_3 = (2,2,2)^\top$. The squared errors are 0.5, 0.5 and 2, with total $3 = \lambda_2$, in agreement with Proposition 4(b). With $k = 2$ the reconstruction is exact, since the rank is 2.

4 Experiment and results

4.1 Data and protocol

The data are 50 frontal photographs of 10 people, five per person, each 180×200 pixels. Every photograph was converted to 8-bit grayscale, kept as double-precision values in $[0, 255]$ without rescaling, and reshaped to a vector in $\mathbb{R}^{36000}$, giving a data matrix of shape $36\,000 \times 50$. After centring, the largest absolute row mean of $\mathbf{A}$ was 1.1×10^{-14}, that is, zero up to rounding.

For each person the photographs were sorted by file name; the first four form the training set and the fifth the test set, so $\mathbf{X}_{\text{train}} \in \mathbb{R}^{36000\times 40}$ and $\mathbf{X}_{\text{test}} \in \mathbb{R}^{36000\times 10}$. The mean face, the Gram matrix, the eigenfaces and the number k were computed from the training images *only*; the test photographs were then treated as new faces and projected with $\boldsymbol{\mu}_{\text{train}}$ and $\mathbf{Q}_k$. This separation matters: if the test images contributed to the basis, they would help to define the space in which they are judged, and the accuracy would be overstated. The computation was done in Python with NumPy; the symmetric eigenproblem for $\mathbf{L}$ used `numpy.linalg.eigh`.

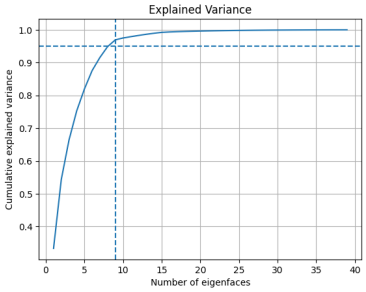
4.2 Eigenfaces and the choice of k

On all 50 images the smallest eigenvalue of $\mathbf{L}$ was -2.76×10^{-8}, about 3×10^{-17} times the largest ($\lambda_1 = 8.69 \times 10^{8}$). This is the rounding image of the zero eigenvalue and was discarded, leaving 49 eigenfaces, the maximum allowed by Lemma 1. The spectrum decays quickly: λ_4 is already

less than a third of λ_1, and $\lambda_{10} = 2.9 \times 10^7$ is about 3% of it. Nine components retain $R_9 = 95.68\%$ of the variance of the 50 images. For the training set the Gram matrix has 39 valid eigenvalues and again $k = 9$ is the smallest number reaching 95%, with $R_9 = 96.94\%$ (Figure 2b). Each face is therefore described by nine coordinates instead of 36 000 pixel values, a reduction of the per-image description by a factor of 4000; the mean face and the nine eigenfaces are common to all images and stored once.



(a) Mean face (50 images)



(b) Cumulative variance R_k, training set

Figure 2. (a) The mean face is blurred, since it averages ten different people; only the coarse structure of head, hair and eyes remains. (b) The dashed lines mark the 95% level and $k = 9$.

4.3 Reconstruction and recognition

Figure 3 compares a photograph with its reconstruction $\hat{\mathbf{x}} = \boldsymbol{\mu} + \mathbf{Q}_9\mathbf{s}$ from nine components of the 50-image decomposition. Overall shape, pose and the outline of the hair are kept, while fine texture such as individual curls is smoothed away, as expected for an approximation that discards 4.3% of the variance (Proposition 4(b)).

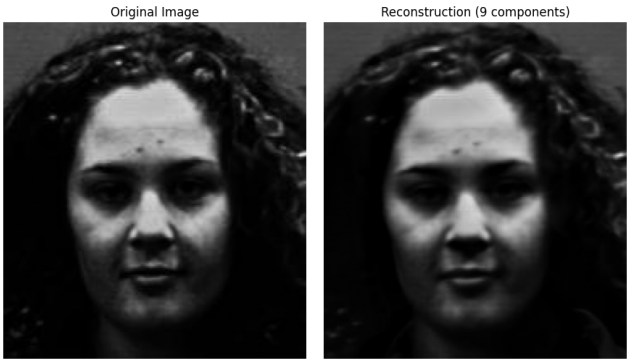


Figure 3. A photograph (left) and its reconstruction from nine eigenfaces (right).

The training and test score matrices have shapes 40×9 and 10×9. Each of the ten test faces was assigned the identity of its nearest training face in the nine-dimensional score space, and all ten assignments were correct (Table 1), an accuracy of 100%.

5 Practical difficulties

Size. The covariance matrix is $36\,000 \times 36\,000$ and cannot be decomposed on an ordinary machine. Solving the 40×40 (or 50×50) problem for $\mathbf{L}$ and mapping back with $\mathbf{u}_i = \mathbf{A}\mathbf{v}_i / \|\mathbf{A}\mathbf{v}_i\|$ gives the same eigenvalues and the eigenfaces at negligible cost (Proposition 3).

Singularity of the Gram matrix. Because $\mathbf{A}\mathbf{1} = \mathbf{0}$, the Gram matrix has a zero eigenvalue, which floating-point arithmetic returns as a small negative number. Taking its square root or dividing

Table 1. Recognition of the ten held-out photographs (actual identity equals predicted identity in every case).

Identity	Correct	Identity	Correct
anpage	✓	pacole	✓
asamma	✓	pspliu	✓
drbost	✓	rsanti	✓
ekavaz	✓	sjbeck	✓
klclar	✓	skumar	✓

by $\|\mathbf{Av}\| \approx 0$ would produce meaningless output, which is why the tolerance $10^{-10}\lambda_1$ is applied. It leaves 49 eigenfaces for the full set and 39 for the training set, exactly the bound of Lemma 1.

Normalisation. The vectors $\mathbf{Av}_i$ have length $\sqrt{\lambda_i}$, not 1. Used as they are, they would inflate the i-th coordinate by $\sqrt{\lambda_i}$; the matrix $\mathbf{Q}$ would not be orthonormal, the scores would not be orthogonal projections, the reconstruction $\boldsymbol{\mu} + \mathbf{Qs}$ would fail, and the distances would weight directions by an uncontrolled factor. Dividing each by its length restores $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}$ and makes the scores genuine coordinates.

6 Limitations and conclusion

Limitations. Ten test images give a coarse assessment. Treating them as independent trials, 10 successes out of 10 are compatible with a wide range of true accuracies: the exact binomial 95% confidence interval is about $[0.69, 1]$. The experiment shows that the method works on this data set and that nine numbers per face suffice here; it does not establish a recognition rate near 100% in general. Cross-validation, rotating which photograph of each person is held out, and a larger collection would give a firmer estimate. The eigenfaces capture overall variance, which may reflect lighting or pose rather than identity, and the data were not designed to test robustness to such changes. With $N \ll d$ the sample covariance has rank at most $N - 1$ and is a poor estimate of the population covariance; it is used here only to select a subspace. Finally, Euclidean distance treats all score coordinates alike, whereas the j-th has variance λ_j/N (Proposition 4); rescaling by $\lambda_j^{-1/2}$ (a Mahalanobis distance) is a natural alternative, and a distance threshold would allow faces of unknown people to be rejected.

Conclusion. The eigenvectors of the scatter matrix of centred face images give the directions along which faces differ, and the eigenvalues give the variance in each. The $36\,000 \times 36\,000$ eigenproblem is replaced by a 40×40 one through the Gram-matrix identity, and the resulting scores are uncorrelated with variances λ_j/N, with the unexplained variance equal to the reconstruction error. On the photographs studied, nine eigenfaces retain about 95% of the variance, and a nearest-neighbour rule on the nine scores identified every held-out face. Given the small test set, we regard this as a successful demonstration of the method rather than a measure of its general accuracy.

References

[1] A. Chakraborty, *Eigen Analysis*, YouTube playlist.